# RED

1. $Fe(SCN)_3 \downarrow$, $Fe(CH_3COO)_3 \downarrow$ — Blood red
2. $Fe(OH)_3 \downarrow$
3. $Ag_2CrO_4 \downarrow$, $Hg_2CrO_4 \downarrow$ — Brick red
4. $Ag_3[Fe(CN)_6] \downarrow$ — orange red
5. $Br_2 + H_2O$
6. $HgI_2 \downarrow$ — Scarlett Red
7. $Cu_2O \downarrow$ — Brick Red
8. $[BiI_4]^-$ — Orange

$I^- \uparrow$ ← $BiI_3 \downarrow$ (Black) ← $Bi^{+3} + KI$

9. $Ni(DMG)_2 \downarrow$ — rosy red
10. $CrO_2Cl_2$ — red volatile liq w/ orange red fumes

# GREEN

1. $Cu_3[Fe(CN)_6]_2 \downarrow$
2. $Cr(OH)_3 \downarrow$ — dirty green
3. $FeCl_2$
4. $Ni(CN)_2 \downarrow$ (BBY GY)
5. $NiCO_3 \downarrow$ (Nigeria)
6. $Na_3[Co(CO_3)_3]$ — YG tile 43→33
7. $Hg_2I_2 \downarrow$ ← $Hg_2^{2+} + I^-$

→ $Hg \downarrow$ (black) $+ [HgI_4]^{2-}$

8. $Na_2MnO_4$

↑ $Na_2CO_3$ / $KNO_3$ → $MnO_2 \cdot H_2O$

* $HMnO_4$ — pink

# BUFF

1. $MnS \downarrow$
2. $Co(CN)_2 \downarrow$ BBY

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# BROWN

1. $Cu_2[Fe(CN)_6]\downarrow$ Chocolate
2. $Bi_2S_3\downarrow$
   $SnS\downarrow$
3. $Fe[Fe(CN)_6]$
4. $K_4[Co(CN)_6]$ BBY
5. $Cu_2I_2\downarrow + I_3^-$
   ↑ white alone
6. $Ag_2O\downarrow$ yellow brown
   ↑ $Ag_2CO_3 \leftarrow CO_3^{2-} + Ag^+$ white
7. $[Fe(H_2O)_5NO]SO_4$ brown ring
8. $HgOHgNH_2I\downarrow \leftarrow NH_4^{\oplus} + {}^{-}OH + K_2[HgI_4]$

# BLUE

1. $Fe_4[Fe(CN)_6]_3\downarrow$ Prussian
   $Fe_3[Fe(CN)_6]_2\downarrow$ Turnbull's
2. $I_2$ in starch
3. $CuSO_4 \cdot Cu(OH)_2\downarrow$
4. $[Cu(NH_3)_2]SO_4$
5. $Cu(OH)_2\downarrow$
6. $Cu_2I_2\downarrow + I_2$ Violet
7. $Na_4[Fe(CN)_5NOS]$ Sodium thio nitroprusside
8. $(NH_4)_2[Co(SCN)_4] \leftarrow Co^{+2} + NH_4SCN$

# BLACK

1. $HgS$, $CoS$, $NiS$, $Ag_2S$, $PbS$, $CuS$ ↓
   ↑ $Ag_2S_2O_3$ (white)
2. $Hg\downarrow + NH_2HgCl\downarrow$ Shining black
3. $Ni_2O_3\downarrow$ (Nigeria)
4. $BiI_3\downarrow$

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